

**School of Arts and Sciences**  
**Department of Computer Science and Information Technology**  
Fall 2025 – 2026

**CSCI342 Fundamentals of Networking Technologies**

|                     |       |                       |       |                      |       |
|---------------------|-------|-----------------------|-------|----------------------|-------|
| <b>Class Day</b> :  | _____ | <b>Class Time</b> :   | _____ | <b>Room</b>          | _____ |
| <b>Instructor</b> : | _____ |                       |       |                      |       |
| <b>E-mail</b> :     | _____ |                       |       |                      |       |
| <b>Office</b> :     | _____ | <b>Office Hours</b> : | _____ | <b>Extension No:</b> | _____ |

## I. Course Description

The course introduces the architecture, structure, functions, components, and models of the Internet and other computer networks. The principles and structure of IP addressing and the fundamentals of Ethernet concepts, media, and operations are introduced to provide a foundation for the CCNA curriculum.

The CCNA® curriculum (version 7.0) helps students enrolled in the Cisco Networking Academy® program prepare for entry-level career opportunities, continuing education with networking related degree programs, and globally recognized Cisco certifications.

The CCNA® includes three courses that make up the recommended course flow: Introduction to Networks, Switching, Routing, and Wireless Essentials Course Resources, and Enterprise Networking, Security, and Automation Course Resources

The CSCI342 course matches with the first course in the flow, Introduction to Networks 7.0 (ITN).

## II. Course Objectives

**CO-1:** Understand Networking Protocols

**CO-2:** Configure Switches and Routers

**CO-3:** Implement IP addressing schemes

**CO-4:** Design and build a simple LAN

### III. Course Learning Outcomes

**CLO – 1:** Explain General Networking Concepts

**CLO – 2:** Apply Basic Configuration of Networking Devices (Switches and Routers)

**CLO – 3:** Classify and Explain Network Protocols and Communication (OSI and TCP/IP)

**CLO – 4:** Design and Create IP Addressing and Subnetting schemes

**CLO – 5:** Analyze and Test basic connectivity issues in simple network topologies

### IV. Mapping to program outcomes

A student who successfully fulfills the course requirements will have demonstrated ability to:

| Fundamentals of Networking Technologies | PO-1 | PO-2 | PO-3 | PO-4 | PO-5 | PO-6 | PO-7 | PO-8 | PO-9 |
|-----------------------------------------|------|------|------|------|------|------|------|------|------|
| <b>CLO – 1</b>                          | x    |      |      |      |      |      |      |      |      |
| <b>CLO – 2</b>                          |      | x    |      | x    |      |      |      |      |      |
| <b>CLO – 3</b>                          | x    |      |      |      |      |      |      |      |      |
| <b>CLO – 4:</b>                         |      |      | x    |      |      |      |      |      |      |
| <b>CLO – 5</b>                          |      | x    |      |      |      |      |      |      |      |

### V. Topics Covered

- Topic I: Introduction to Networks
- Topic II: The CCNA LAB Environment
- Topic III: Network Layered Models and Protocols
- Topic IV: Physical & Data Link Layer Protocols
- Topic V: Network Layer
- Topic VI: Transport Layer
- Topic VII: Application Layer

### VI. Course Prerequisite(s)

CSCI250 Introduction to Programming

### Course Corequisite(s)

None

### VII. Course Material: Book or handouts

Introduction to Networks Companion Guide (CCNAv7).

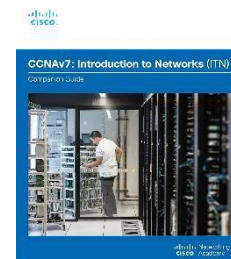
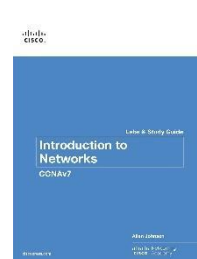
**Publisher:** Cisco Press; 1 edition (July 14, 2020)

**ISBN-10:** 0136633668

Introduction to Networks Labs and Study Guide

(CCNAv7) (Lab Companion). **Publisher:** Cisco Press; 1

edition (June 17, 2020) **ISBN-10:** 0136634451



### VIII. Software required (for computer-lab courses)

- Internet browser
- CISCO packet tracer > 8.2.2

## IX. Online Cisco Assessment

### Grading Policy and Withdrawal Deadline

|                   |                    |     |     |                                       |
|-------------------|--------------------|-----|-----|---------------------------------------|
| LIU Exams         | Midterm Exam       |     | 25% | <i>To be announced</i>                |
|                   | Final Exam         |     | 35% | <i>To be announced</i>                |
| CISCO Assessments | In-LAB Skills Exam | 25% | 40% | Last Session                          |
|                   | CISCO assignment   | 10% |     | The day after the end of the semester |
|                   | CISCO modules Exam | 5%  |     | During the semester                   |

Withdrawal Deadline: Week 14

## X. Detailed Schedule

| Week   |           | Book Reference                                              | Topic Details                                                                                                                                                                                                                                                                                                 |
|--------|-----------|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 1 | Lecture 1 | Introduction+ <b>Module 1</b> : The network today           | Course syllabus, Explain how networks affect our daily lives. Explain how host and network devices are used. Explain network representations and how they are used in network topologies. Compare the characteristics of common types of networks. Explain how LANs and WANs interconnect to the internet.    |
|        | Lecture 2 | Continue Module 1                                           | Describe the four basic requirements of a reliable network. Explain how trends such as BYOD, online collaboration, video, and cloud computing are changing the way we interact. Identify some basic security threats and solution for all networks. Explain employment opportunities in the networking field. |
| Week 2 | Lecture 1 | <b>Module 2</b> : Basic Switch and End Device Configuration | Introduction to packet tracer<br>Explain how to access a Cisco IOS device for configuration purposes                                                                                                                                                                                                          |
|        | Lecture 2 | Continue Module 2                                           | Explain how to navigate Cisco IOS to configure network devices.<br>Describe the command structure of Cisco IOS software.<br>Configure a Cisco IOS device using CLI.                                                                                                                                           |
| Week 3 | Lecture 1 | Continue Module 2                                           | Use IOS commands to save the running configuration. Explain how devices communicate across network media. Configure a host device with an IP address.<br><i>2.5.5 Packet Tracer - Configure Initial Switch Settings</i>                                                                                       |
|        | Lecture 2 | Continue Module 2                                           | <i>2.7.6 Packet Tracer - Implement Basic Connectivity</i><br><i>2.9.1 Packet Tracer - Basic Switch and End Device Configuration</i>                                                                                                                                                                           |
| Week 4 | Lecture 1 | <b>Module 3</b> : Protocols and Models                      | Describe the types of rules that are necessary to successfully communicate.                                                                                                                                                                                                                                   |

|                            |           |                                              |                                                                                                                                                                                                                                                                                                                                                                                      |
|----------------------------|-----------|----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| k<br>4                     | Lecture 2 |                                              | <p>Explain why protocols are necessary in network communication.</p> <p>Explain the purpose of adhering to a protocol suite.<br/>Explain the role of standards organizations in establishing protocols for network interoperability.</p>                                                                                                                                             |
|                            |           | Continue Module 3                            | <p>Explain how the TCP/IP model and the OSI model are used to facilitate standardization in the communication process.<br/>Explain how data encapsulation allows data to be transported across the network.<br/>Explain how local hosts access local resources on a network.</p>                                                                                                     |
| W<br>e<br>e<br>k<br>5      | Lecture 1 | <b>Module 4:</b> Physical layer              | <p>Describe the purpose and functions of the physical layer in the network.<br/>Describe characteristics of the physical layer.<br/>Identify the basic characteristics of copper cabling.<br/>Explain how UTP cable is used in Ethernet networks.<br/>Describe fiber optic cabling and its main advantages over other media.<br/>Connect devices using wired and wireless media.</p> |
|                            | Lecture 2 | <b>Module 6:</b> Data link layer             | <p>Describe the purpose and function of the data link layer in preparing communication for transmission on specific media.</p>                                                                                                                                                                                                                                                       |
| W<br>e<br>e<br>k<br>s<br>6 | Lecture 1 | Continue module 6                            | <p>Compare the characteristics of media access control methods on WAN and LAN topologies.<br/>Describe the characteristics and functions of the data link frame.</p>                                                                                                                                                                                                                 |
|                            | Lecture 2 | <b>Module 7:</b> Ethernet Switching          | <p>Explain how the Ethernet sublayers are related to the frame fields.<br/>Describe the Ethernet MAC address.<br/>Explain how a switch builds its MAC address table and forwards frames.<br/>Describe switch forwarding methods and port settings available on Layer 2 switch ports.<br/>Exercise on MAC address building and maintaining</p>                                        |
| W<br>e<br>e<br>k<br>7      | Lecture 1 | <b>Module 8:</b> Network Layer               | <p>Explain how the network layer uses IP protocols for reliable communications.<br/>Explain the role of the major header fields in the IPv4 packet.</p>                                                                                                                                                                                                                              |
|                            | Lecture 2 | Continue module 8                            | <p>Explain how network devices use routing tables to direct packets to a destination network.<br/>Explain the function of fields in the routing table of a router</p>                                                                                                                                                                                                                |
| W<br>e<br>e<br>k<br>8      | Lecture 1 | <b>Module 9:</b> Address Resolution          | <p>Compare the roles of the MAC address and the IP address.<br/>Describe the purpose of ARP.</p>                                                                                                                                                                                                                                                                                     |
|                            | Lecture 2 | <b>Module 10:</b> Basic Router Configuration | <p>Configure initial settings on an IOS Cisco router.<br/>Configure two active interfaces on a Cisco IOS router.<br/><i>10.3.4 Packet Tracer - Connect a Router to a LAN</i></p>                                                                                                                                                                                                     |
| W<br>e<br>e<br>k<br>9      | Lecture 1 | Continue module 10                           | <p>Configure devices to use the default gateway.<br/><i>10.1.4 Packet Tracer – Configure initial router settings</i></p>                                                                                                                                                                                                                                                             |
|                            | Lecture 2 | <b>Module 11:</b> IPv4 Addressing            | <p>Describe the structure of an IPv4 address including the network portion, the host portion, and the subnet mask.</p>                                                                                                                                                                                                                                                               |

|         |                        |                                     |                                                                                                                                                                                                                                                                                                 |
|---------|------------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|         |                        |                                     | Compare the characteristics and uses of the unicast, broadcast and multicast IPv4 addresses.                                                                                                                                                                                                    |
| Week 10 | Lecture 1              | Continue module 11                  | Explain public, private, and reserved IPv4 addresses.<br>Explain how subnetting segments a network to enable better communication.                                                                                                                                                              |
|         | Lecture 2              | Continue module 11                  | Equal-sized subnetting<br>Exercises on equal size subnetting                                                                                                                                                                                                                                    |
| Week 11 | Lecture 1              | Continue module 11                  | VLSM<br>Exercises on vlsn                                                                                                                                                                                                                                                                       |
|         | Lecture 2              | Continue module 11                  | Extra exercises on vlsn and equal size subnetting                                                                                                                                                                                                                                               |
| Week 12 | Lecture 1              | <b>Module 14:</b> Transport Layer   | Explain the purpose of the transport layer in managing the transportation of data in end-to-end communication.<br>Explain characteristics of TCP.<br>Explain characteristics of UDP.<br>Explain how TCP and UDP use port numbers                                                                |
|         | Lecture 2              | Continue module 14                  | Explain how TCP session establishment and termination processes facilitate reliable communication.<br>Explain how TCP protocol data units are transmitted and acknowledged to guarantee delivery.<br>Compare the operations of transport layer protocols in supporting end-to-end communication |
| Week 13 | Lecture 1              | <b>Module 15:</b> Application Layer | Explain how web and email protocols operate.<br>Explain how DNS and DHCP operate.<br>Explain how file transfer protocols operate.                                                                                                                                                               |
|         | Lecture 2              | DAY OFF                             |                                                                                                                                                                                                                                                                                                 |
| Week 14 | Lecture 1              | DAY OFF                             |                                                                                                                                                                                                                                                                                                 |
|         | Lecture 2              | DAY OFF                             |                                                                                                                                                                                                                                                                                                 |
| Week 15 | Lecture 1<br>Jan 6 OFF | Revision                            |                                                                                                                                                                                                                                                                                                 |
|         | Lecture 2              | Skills exam                         |                                                                                                                                                                                                                                                                                                 |

### Instructions for Exams

- Cheating in any way, form or method will be rewarded with an F grade.
- You should read each question carefully before answering.
- You should solve the questions that you are confident about their answers first.
- Calculators not allowed.
- Neither an Open Book nor an Open Notes exam.
- Laptop and USB not allowed.

## XI. Class Rules & Discipline

### Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

### Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the LIU student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

### Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

### Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

### Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

### Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam's grade. The percentage % will be set by the administration once the petition is approved.

**Cheating policy**: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

### Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

### Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

### Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

### LIU E-mails

LIU students should use their LIU e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your LIU e-mail accounts.

### University Peer Review Policy

This is a new policy at LIU in which during the semester, all classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

### Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

## Equivalent Lettering Grades

| <u>Percent (%)</u> | <u>QPTS</u> | <u>Letter Grade (G)</u> | <u>Percent (%)</u> | <u>QPTS</u> | <u>Letter Grade (G)</u> |
|--------------------|-------------|-------------------------|--------------------|-------------|-------------------------|
| <60                | 0           | F                       | 67                 | 1.7         | C                       |
| 60                 | 1           | D                       | 68                 | 1.8         |                         |
| 61                 | 1.1         |                         | 69                 | 1.9         |                         |
| 62                 | 1.2         |                         | 70                 | 2           |                         |
| 63                 | 1.3         |                         | 71                 | 2.1         |                         |
| 64                 | 1.4         | D+                      | 72                 | 2.2         | B+                      |
| 65                 | 1.5         |                         | 73                 | 2.3         |                         |
| 66                 | 1.6         |                         | 74                 | 2.4         |                         |
| 75                 | 2.5         | C+                      | 83                 | 3.3         |                         |
| 76                 | 2.6         |                         | 84                 | 3.4         |                         |
| 77                 | 2.7         |                         | 85                 | 3.5         |                         |
| 78                 | 2.8         | B                       | 86                 | 3.6         | A                       |
| 79                 | 2.9         |                         | 87                 | 3.7         |                         |
| 80                 | 3           |                         | 88                 | 3.8         |                         |
| 81                 | 3.1         |                         | 89                 | 3.9         | A                       |
| 82                 | 3.2         |                         | >=90               | 4           |                         |

| <u>Grade</u> | <u>Description</u>                        | <u>QPTS</u> |
|--------------|-------------------------------------------|-------------|
| A            | Excellent                                 | 4.00        |
| B+           | Very Good                                 | 3.50        |
| B            | Good                                      | 3.00        |
| C+           | Above Average                             | 2.50        |
| C            | Average                                   | 2.00        |
| D+           | Fair                                      | 1.50        |
| D            | Poor                                      | 1.00        |
| F            | Fail                                      | 0.00        |
| IP           | Course in Progress                        |             |
| I            | Incomplete work                           |             |
| W            | Student has withdrawn from the course     |             |
| AW           | Administrative withdrawal from the course |             |
| R            | Course has been repeated                  |             |
| S            | Satisfactory work                         |             |
| U            | Unsatisfactory work                       |             |
| %G           | Final Grade                               |             |
| QPTS         | Quality Points                            |             |

## XIII. Computer sciences outcomes

|                                                                               |                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Program Outcomes</b>                                                       | <b>Computer Science Outcomes</b><br>Graduates of the Computer Science Program will have the ability to:                                                                                                                                                                                    |
| <b>category -1 Underlying Conceptual Basis for Informatics</b>                | <b>PO-1 describe</b> and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study                                                 |
|                                                                               | <i>PO-1 for CSIT=&gt; Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>                                                                                                              |
| <b>category -2 Analysis</b>                                                   | <b>PO-2 select</b> and use relevant analytic, modelling and simulation methods                                                                                                                                                                                                             |
|                                                                               | <b>PO-3</b> choose appropriate solution patterns, algorithms and data structures                                                                                                                                                                                                           |
| <b>category -3 Design and Implementation</b>                                  | <b>PO-4 describe</b> the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts                                                                                                         |
|                                                                               | <i>PO-4 for CSIT=&gt;specify and design computing/network hardware/software which meet specified requirements</i>                                                                                                                                                                          |
| <b>category -4 Economic, legal, social, ethical and environmental context</b> | <b>PO-5</b> explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems                                                                                                         |
| <b>category -5 Informatics practice</b>                                       | <b>PO-6</b> describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques |
|                                                                               | <i>PO-6 for CSIT=&gt; demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>                                                                                                                     |
| <b>category -6 Other Professional Skills and Competences</b>                  | <b>PO-7 communicate</b> effectively both verbally and using a variety of communications media                                                                                                                                                                                              |
|                                                                               | <b>PO-8</b> plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development                                                                                                                                                 |
|                                                                               | <b>PO-9 participate</b> effectively in informatics group-working                                                                                                                                                                                                                           |



# CSCI342

